TECHNICAL REPORT

Customer Churn Analysis | Week 8 Capstone Project

1. Executive overview

This report documents an end-to-end analysis of the customer churn dataset supplied for the internship capstone. The dataset has 500 observations and 9 fields. The project focuses on data understanding, quality checks, descriptive analysis, segmentation, visualisation, and business recommendations.

2. Business context and objectives

The business problem is to understand customer loss and identify actionable patterns that may help improve retention. The analysis objectives are to quantify churn, compare churn across available customer segments, explore numerical customer attributes, and communicate evidence-based recommendations.

3. Dataset overview

Field	Data type	Example / note
CustomerID	object	C00001
Tenure	int64	6
MonthlyCharges	int64	64
TotalCharges	int64	1540
Contract	object	One year
PaymentMethod	object	Credit Card
PaperlessBilling	object	No
SeniorCitizen	int64	1
Churn	int64	0

Data quality checks found 0 missing cells and 0 duplicate rows in the original file. Confirm the meaning and encoding of each field before using results for business decisions.

4. Data preparation

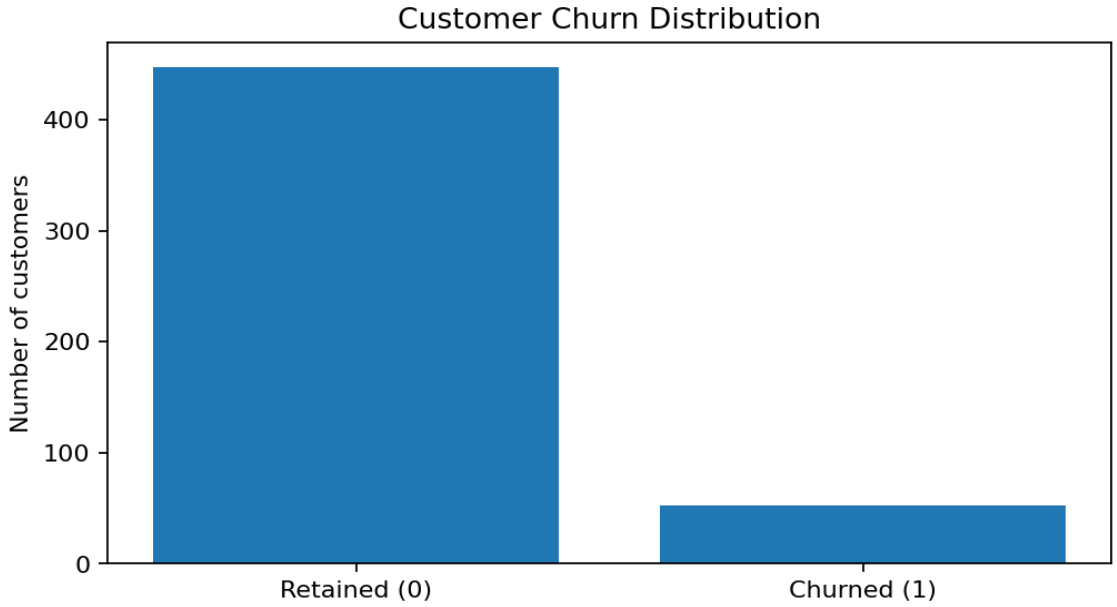
The raw CSV is preserved unchanged. A separate cleaned dataset is created by removing duplicate rows and handling missing values with a column-appropriate approach (median for numeric fields and mode for categorical fields). In this supplied file, the quality checks indicate whether such replacements are necessary. Any imputation should be documented and reviewed before modelling.

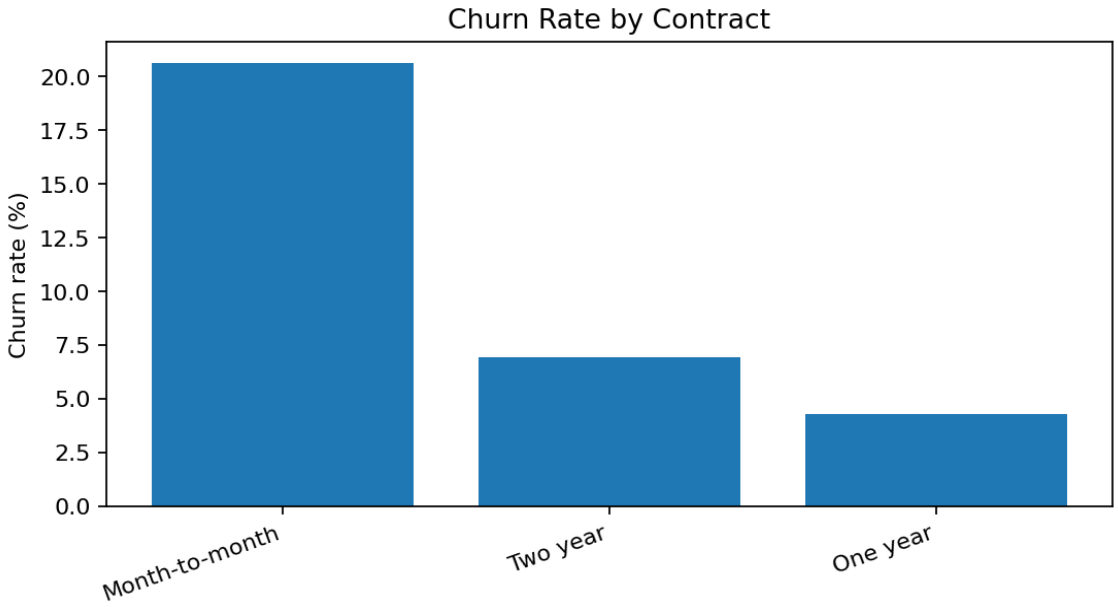
5. Analytical methods

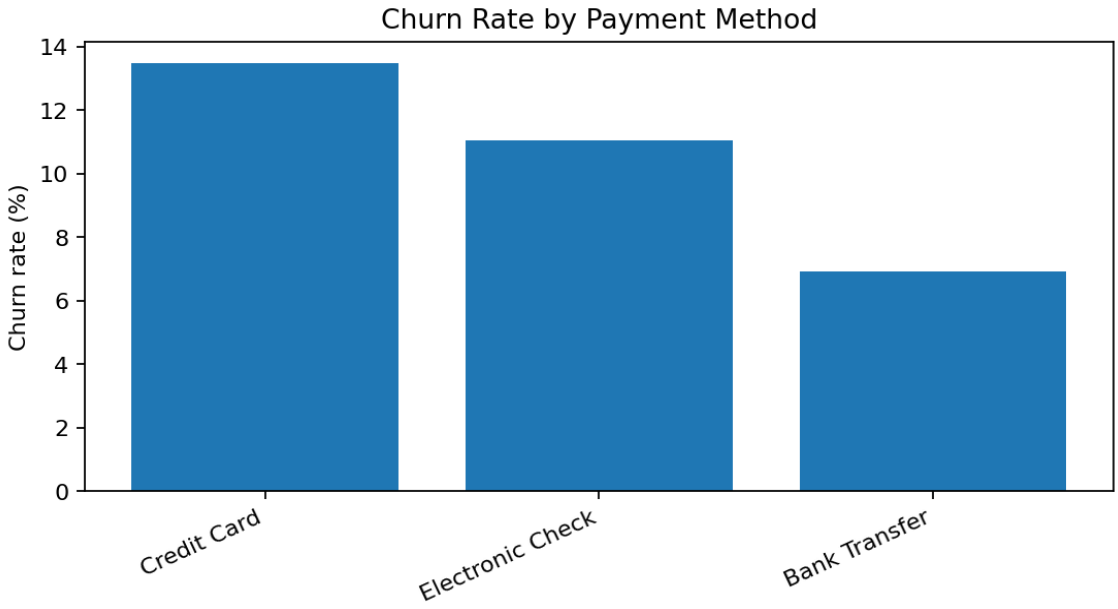
Descriptive analysis: row counts, distributions, averages, medians, ranges and churn rate. **Segmentation:** group customers by contract and payment method (if present) and compare within-group churn rates. **Relationship analysis:** inspect numerical correlations and compare tenure/charges across churn outcomes. **Predictive analysis:** a logistic-regression baseline can be used when the target and predictor fields are suitable; use a stratified train/test split and evaluate precision, recall, F1-score and confusion matrix.

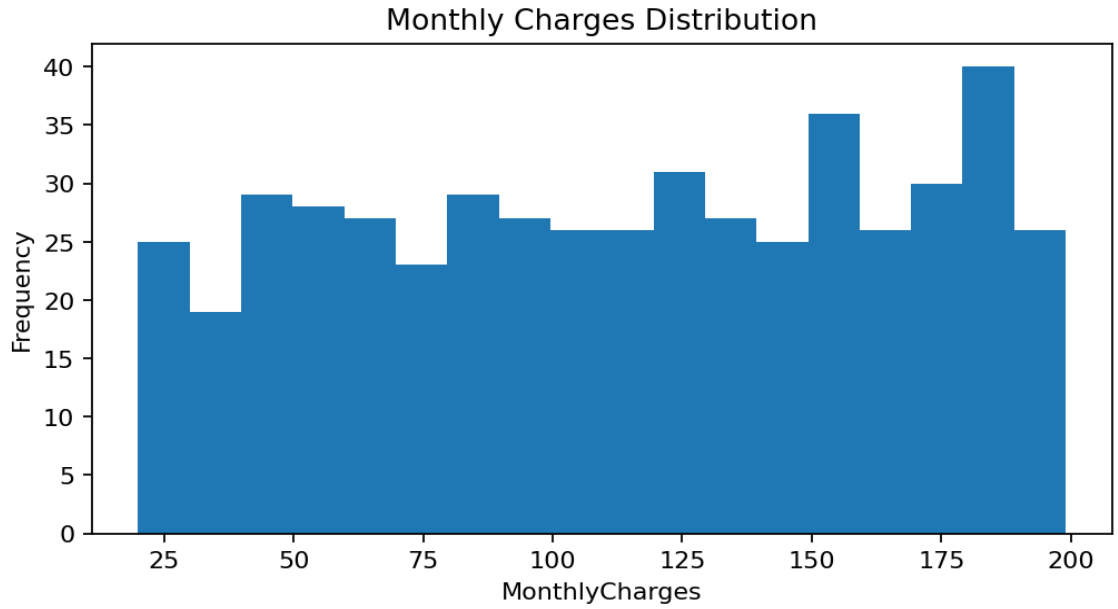
6. Exploratory data analysis and visualisations

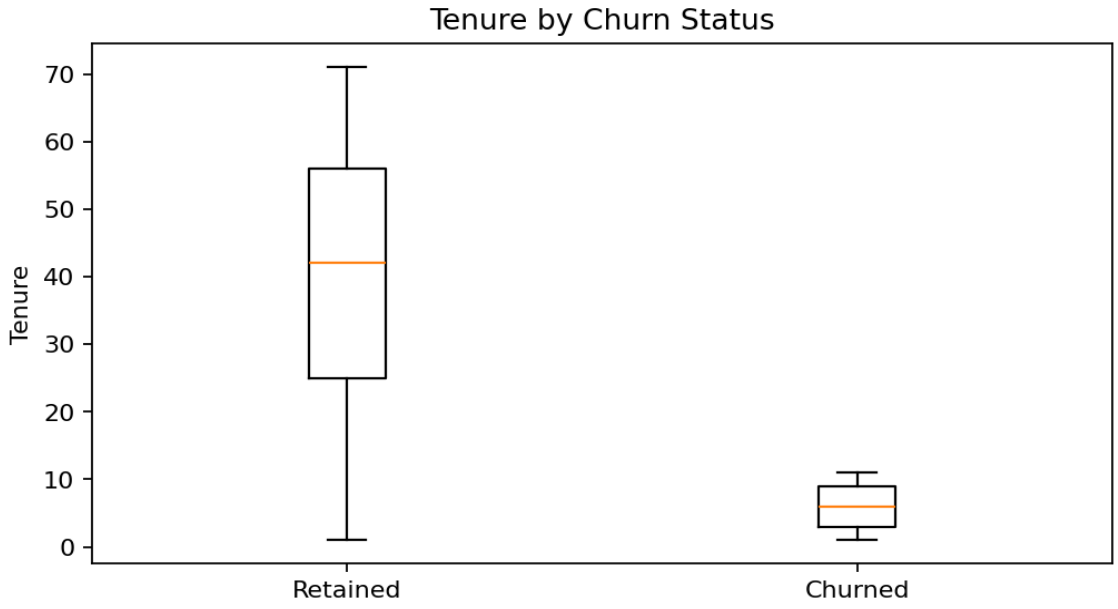
The charts below are generated directly from the supplied CSV. Each chart should be interpreted alongside sample size and segment definitions. Differences between groups are descriptive and do not establish causation.

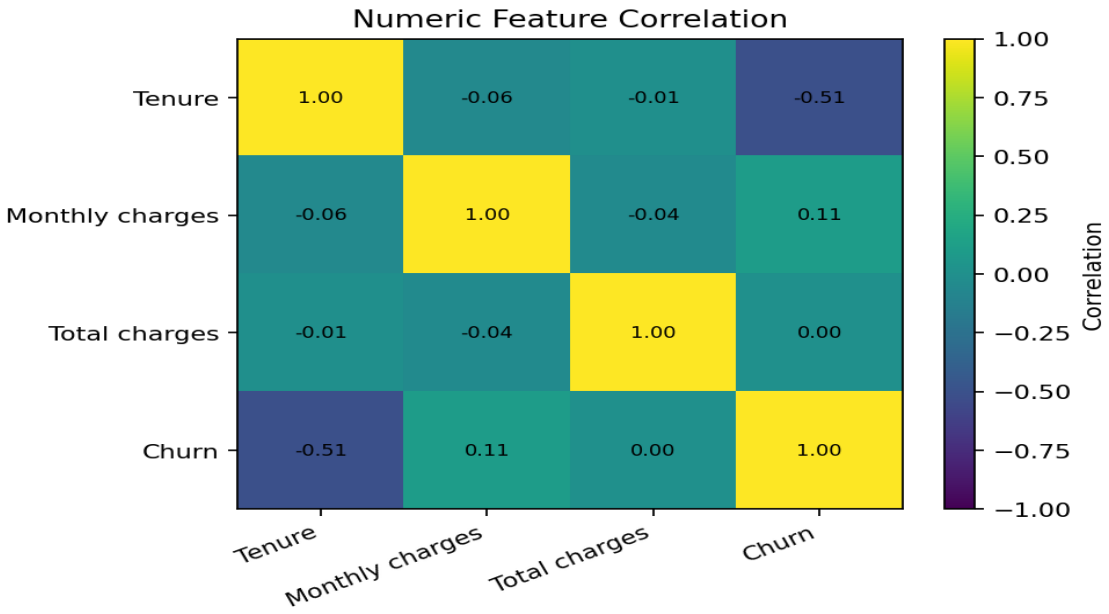





Churn Rate by Payment Method
Churn rate (%)
0
2
4
6
8
10
12
14
Credit Card
Electronic Check
Bank Transfer


Monthly Charges Distribution
Frequency
0
5
10
15
20
25
30
35
40
25
50
75
100
125
150
175
200
MonthlyCharges


Tenure by Churn Status
Tenure
0
10
20
30
40
50
60
70
Retained
Churned



Contract segment detail

• Month-to-month: 20.6% churn

• Two year: 6.9% churn

• One year: 4.3% churn

Payment method segment detail

• Credit Card: 13.5% churn

• Electronic Check: 11.0% churn

• Bank Transfer: 6.9% churn

7. Model evaluation and validation

If a logistic-regression model is included, encode categorical predictors within a reproducible pipeline, split the data using stratification, and avoid including identifiers as predictors. Report the confusion matrix and class-specific precision/recall. Because churn classes can be imbalanced, accuracy alone may be misleading. Validate probability thresholds against the business cost of missed churners versus unnecessary retention outreach.

8. Business recommendations and implementation plan

1. Segment-led retention: target customer groups with higher observed churn and test tailored retention messages. **2. Early-tenure monitoring:** identify onboarding friction and proactively contact new customers. **3. Contract experience:** test clear plan comparisons and suitable incentives for longer commitments. **4. Payment experience:** investigate friction in payment methods with higher churn. **5. Measurement:** track churn rate, retention rate, campaign response, and customer lifetime value monthly.

Suggested implementation: establish a baseline; select one segment; design a small controlled pilot; compare treatment and control outcomes; calculate incremental retention and campaign cost; then scale only if results are positive.

9. Limitations and reproducibility

The dataset is a limited sample and may omit important factors such as customer satisfaction, support interactions, competitor offers, acquisition channel and time-based behaviour. Correlation is not causation. Keep the original CSV, cleaned CSV, notebooks, library versions and generated outputs together. Re-run notebooks from top to bottom to reproduce results.

10. Conclusion

This capstone demonstrates a complete analytical workflow from raw data inspection to business recommendations. The strongest next step is to validate the descriptive patterns with additional time-period data and controlled retention experiments. Final operational decisions should be based on validated findings and business context.